Phase 2 - Week 5 - Lesson Day 3

Logistic Regression

Today's Goal

Yesterday you predicted a number (price). Today you predict a **category** instead - like yes/no, pass/fail, or spam/not spam. This is called **classification**, and Logistic Regression is the simplest, most common model for it.

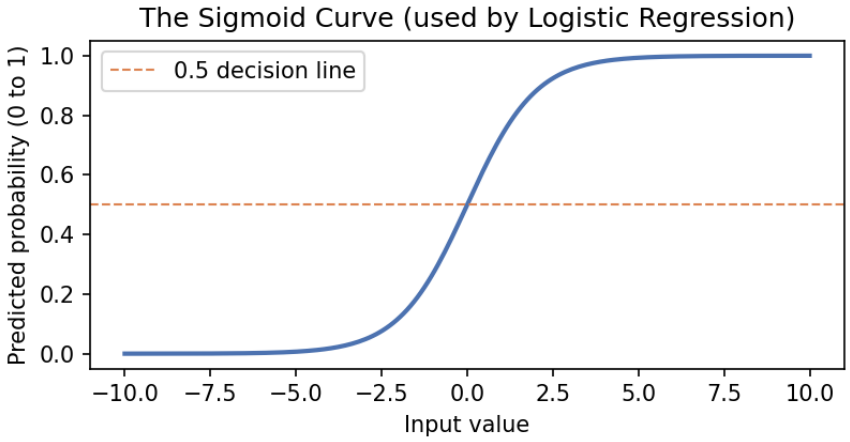
- Understand the difference between regression and classification
- Understand the sigmoid curve
- Train a logistic regression model and classify new data

1. Regression vs Classification

Regression (Day 2) predicts a number - like a price, a temperature, or a score. **Classification** (today) predicts a category - like pass/fail, yes/no, or which group something belongs to. The name "Logistic Regression" is a bit confusing because it has the word "regression" in it, but it is actually used for classification.

2. The Sigmoid Curve

Simple definition: Logistic Regression doesn't directly output "yes" or "no." It first calculates a **probability** between 0 and 1 using a special S-shaped curve called the **sigmoid**. If the probability is above 0.5, it predicts "yes" (or category 1). If it's below 0.5, it predicts "no" (or category 0).



Notice the curve never goes below 0 or above 1 - it always stays in that range, which is exactly what you need to represent a probability. The orange dashed line at 0.5 is the normal cutoff point used to decide between the two categories.

3. Training a Logistic Regression Model

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

# X = input (e.g. hours studied), y = target (0 = fail, 1 = pass)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42
)

model = LogisticRegression()
model.fit(X_train, y_train)   # same .fit() step as Linear Regression

predictions = model.predict(X_test)    # predicted categories (0 or 1)
probabilities = model.predict_proba(X_test)   # the actual probabilities behind each pre
diction
```

Notice this uses the exact same **.fit()** and **.predict()** pattern as Linear Regression yesterday. Scikit-learn deliberately keeps this pattern the same across almost every model, so once you've learned it once, you can apply it to dozens of different models.

.predict_proba() is new - it shows you the actual probability behind each prediction, not just the final yes/no answer. This is useful when you want to know how confident the model is, not just what it guessed.

Today's Practice

Open **day-3-logistic-regression.ipynb**. All answers are already solved.

Practice Questions (5 total - all solved)

1. Build a classification dataset

Create a made-up dataset: hours_studied -> pass/fail (0 or 1).

2. Split and train

Split the data into train/test, then train a LogisticRegression model.

3. Predict categories

Predict pass/fail for the test data and print the results next to the actual values.

4. Predict probabilities

Use .predict_proba() on the test data and print the probabilities for the first 5 students.

5. Score the model

Check the model's accuracy using .score() and print it.

Once you and Halimo finish reviewing today's practice, move to Lesson Day 4 (Comparing Models + Week 5 Mini Challenge).